

1. 2012Q4

Which of the following gives the correct direction of impulse transmission in a neurone?

- A. axon → cell body → dendrites
- B. dendrites → cell body → axon
- C. cell body → axon → dendrites
- D. dendrites → axon → cell body

2. 2012Q5

Chemicals released at the synapse are responsible for the impulse transmission from

- (1) sensory neurone to interneurone.
- (2) interneurone to motor neurone.
- (3) motor neurone to muscle.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

3. 2012Q6

Which of the following activities is coordinated by the cerebellum?

- A. muscular movements during dancing
- B. breathing movements during sleeping
- C. secretion of digestive juices when one is hungry
- D. withdrawal of hand when one touches a hot object

4. 2015Q22

Toys are often used to develop children's fine motor skills. Which of the following parts is trained when children play with toys?

- A. cerebellum
- B. spinal cord
- C. pituitary
- D. medulla oblongata

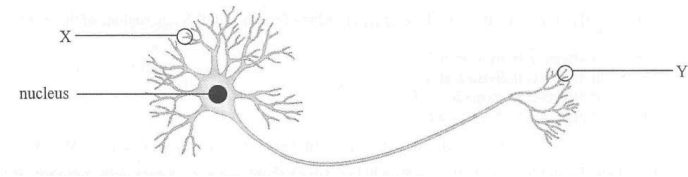
5. 2015Q23

Which of the following correctly compares reflex actions and voluntary actions?

- | | <i>Reflex actions</i> | <i>Voluntary actions</i> |
|----|------------------------------|------------------------------------|
| A. | responses may vary | responses are always the same |
| B. | stimulus is optional | stimulus is required |
| C. | initiated by receptors | initiated in the brain |
| D. | effectors must be muscles | effectors can be muscles or glands |

6. 2017Q29

The diagram below shows a motor neurone:



Which of the following statements correctly describes the motor neurone?

- A. X is connected to a muscle fibre.
- B. Electrical impulses are transmitted from Y to X.
- C. Synapses can be found at the end of both X and Y.
- D. Electrical impulses are sent out at Y to the next neurone.

7. 2017Q30

Which of the following **does not** belong to the central nervous system?

- A. medulla oblongata
- B. spinal cord
- C. cerebellum
- D. eye

8. 2018Q20

Which of the following combinations correctly identifies the types of neurones to which X and Y belong?

- | | <i>X</i> | <i>Y</i> |
|----|-----------------|-----------------|
| A. | (1) | (3) |
| B. | (2) | (1) |
| C. | (2) | (3) |
| D. | (3) | (2) |

9. 2018Q21

Another neural pathway allows the man to feel the tapping action. Which of the following parts should this pathway connect to?

- A. cerebellum
- B. spinal cord
- C. cerebral cortex
- D. medulla oblongata

10. 2018Q22

Which of the following combinations correctly identifies the distribution of grey matter and white matter in the cerebrum and spinal cord?

- | | <i>Inner part of the cerebrum</i> | <i>Inner part of the spinal cord</i> |
|----|--|---|
| A. | white matter | grey matter |
| B. | white matter | white matter |
| C. | grey matter | grey matter |
| D. | grey matter | white matter |

11. 2019Q18

Which of the following organs serves both endocrine and exocrine functions?

- A. pancreas
- B. pituitary
- C. oesophagus
- D. adrenal gland

12. 2019Q21

Daisy is very hungry. When the waiter puts her favourite dish on the table, her saliva secretion increases.

Which of the following body parts controls this response?

- A. cerebrum
- B. cerebellum
- C. salivary gland
- D. medulla oblongata

13. 2020Q10

Which of the following explain why a person cannot swallow food and talk at the same time?

- (1) The epiglottis covers the opening of the trachea during swallowing.
 - (2) Swallowing is a reflex action while speaking is a voluntary action.
 - (3) Air must flow through the larynx to produce sounds.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

14. 2023Q20

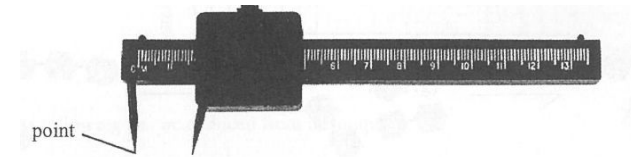
Which of the following correctly explain(s) why reflex actions are usually faster than voluntary actions?

- (1) Reflex actions do not involve decision making.
 - (2) Nerve impulses travel faster in reflex actions.
 - (3) Reflex actions are inborn.
- A. (1) only
 - B. (2) only
 - C. (1) and (3) only
 - D. (2) and (3) only

15. 2024Q23

Questions 23 to 25 refer to a two-point discrimination test on the human skin by following the procedure shown below:

- 1. Blindfold the subjects (tested persons).
- 2. Set the calipers (shown below) to a distance of 30 mm. Apply the two points of the calipers with equal pressure on the fingertip of the subject's index finger. Have the subjects identify if they feel one touch or two touches.



- 3. Repeat step 2 by using a progressively shorter distance (e.g. 28 mm, 26 mm, 24 mm) between the two points until the subject feels one touch. Record the distance.
- 4. Repeat steps 2 and 3 by touching the palm of the hand, the back of the hand and the upper arm.
- 5. Over the course of the experiment, randomly use the caliper set at 0 mm for every couple of trials. It is important to make sure that the subjects tell you that they only feel one touch when the caliper is set at 0mm.

Based on the procedure given, how many independent variable(s) is/are being studied in this investigation?

- A. 1
- B. 2
- C. 3
- D. 4

16. 2024Q24

The investigation was designed to show the density of the touch receptors of the skin. What is the assumption of this investigation?

- A. The subjects do not cheat by peeking out of the blindfold.
- B. The fingertip is more sensitive than the other parts of the body.
- C. A closer distance means that the density of the touch receptors is higher.
- D. The two-touch sensations are produced by two separated touch receptors without overlapping.

17. 2024Q25

Which of the following correctly explains the importance of Step 5 in this investigation?

- A. It ensures the validity of the measurement.
- B. It checks the accuracy of the measurement.
- C. It improves the reliability of the measurement.
- D. It reduces the random error of the measurement.

18. 2024Q26

Which of the following parts of the brain are responsible for coordination when one is speaking?

- (1) cerebellum
 - (2) motor area
 - (3) sensory area
 - (4) association area
- A. (2) and (4) only
 - B. (1), (2) and (3) only
 - C. (1), (2) and (4) only
 - D. (1), (3) and (4) only

LQ

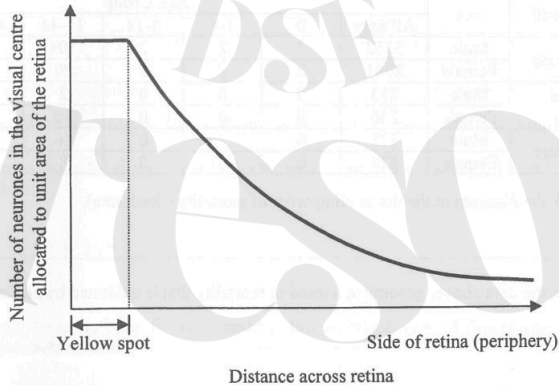
1. 2013Q1

For each of the brain parts listed in column 1, select from column 2 one phrase that matches (3 marks)
it. Put the appropriate letter in the space provided.

Column 1		Column 2
Cerebellum	_____	A. Damage causes inability to make decisions
Medulla oblongata	_____	B. Damage causes loss of body balance
Cerebrum	_____	C. Damage causes loss of control of breathing rate
	_____	D. Damage causes loss of withdrawal reflex

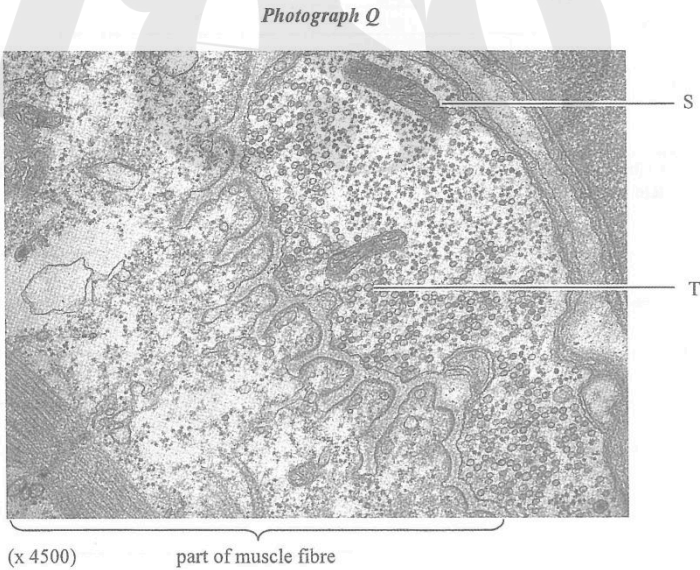
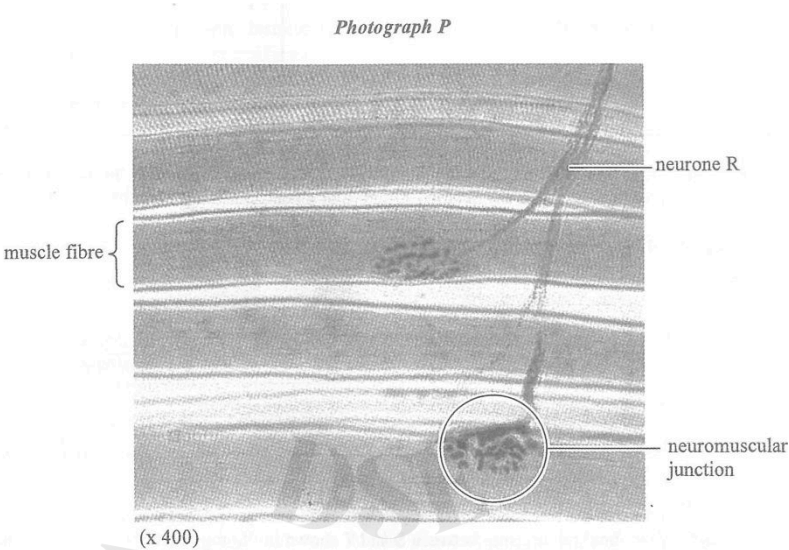
2. 2014Q10b

The visual center of the brain is responsible for processing nerve impulses from the retina. The number of neurones in the visual centre allocated to receiving impulses from a unit area of the retina is shown below (only half of the retina is shown):



b) After the perception of what we ‘see’, what it means to us depends on other parts of the brain. Explain how this works. (2 marks)

3. 2015Q5



Photograph P shows several muscle fibres associated with the terminal parts of neurone R and Photograph Q shows the detailed structure of a neuromuscular junction.

- a) Which type of neurones does R belong to? Give a reason for your answer. (2 marks)
- b) What is the functional relationship between S and T shown in Photograph Q? (2 marks)
- c) Describe how nerve impulses can be transmitted across the neuromuscular junction leading to muscle contraction. (3 marks)

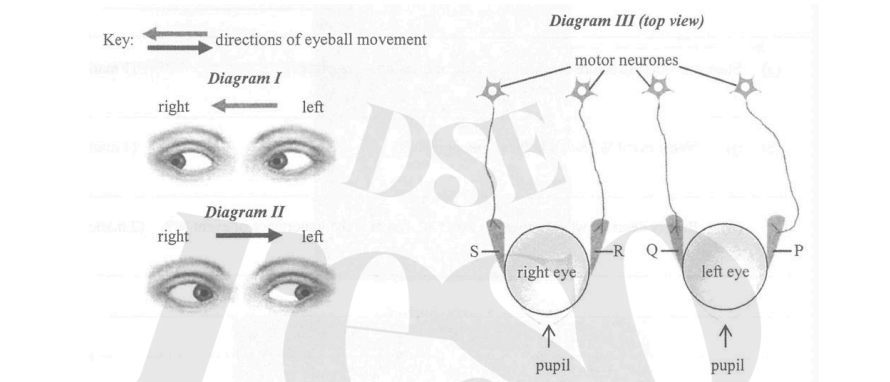
4. 2016Q1

The table below shows the condition which may result from damage to a certain part of the brain. Complete the table by filling in either the condition or the structural part of the brain affected. (3 marks)

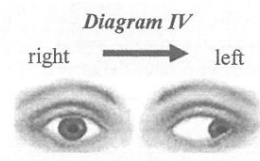
Condition	Structural part of the brain
difficulty in breathing	
	cerebellum
difficulty in speech and vision	

5. 2019Q4

Diagrams I and II below show a person with both eyes moving right and then left. This eyeball movement is brought about by the coordination of different eye muscle pairs. Diagram III shows four of the muscles (P, Q, R, and S), all connected to motor neurons controlling eyeball movement.



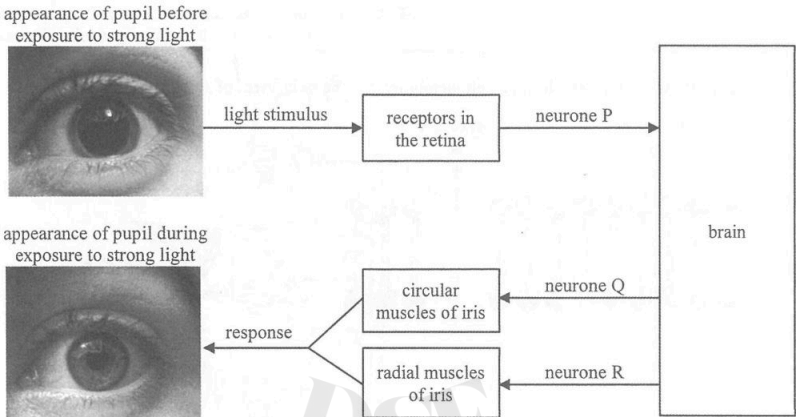
- a) To bring about the eyeball movement shown in Diagram I, which muscle(s) (P, Q, R, or S) contract(s)? (1 mark)
- b) A person suffers from impaired eyeball movement when turning his eyes from right to left, as shown in Diagram IV.



It is found that one of his eye muscles cannot fully contract. Based on your knowledge of neurotransmission at the neuromuscular junction, suggest **two** possible defects that would lead to the impaired eyeball movement shown in Diagram IV. (2 marks)

6. 2020Q3aii, c

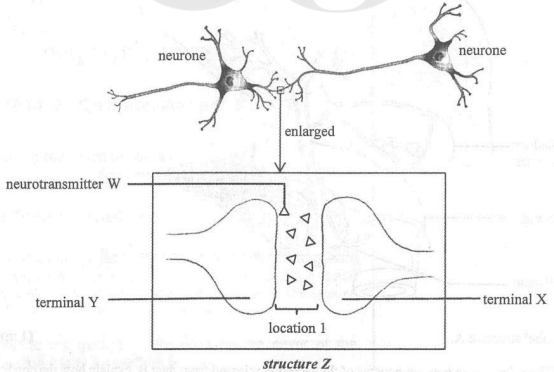
The diagram below shows a neural pathway involved in the coordination of pupil size when a human eye is exposed to strong light:



- a) With reference to the above neural pathway,
- ii) state the type of neurone represented by Q and R. (1 mark)
- c) If someone falls unconscious, the response of the above neural pathway will be assessed to confirm if the pathway is still functioning. What does this assessment tell you about the nature of this neural coordination? (1 mark)

7. 2021Q2

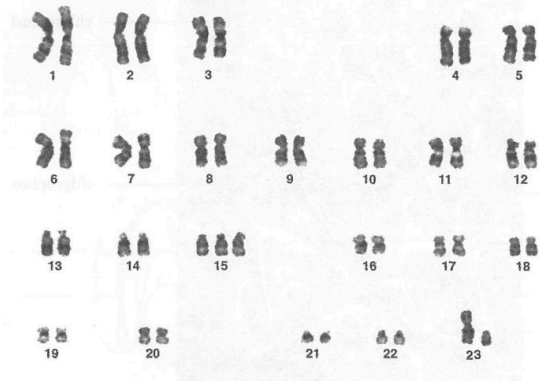
The diagram below shows two adjacent neurones. When a nerve impulse arrives at structure Z, the amount of neurotransmitter W at location 1 increases.



- a) Name structure Z. (1 mark)
- b) i) Neurotransmitter W at location 1 is released from one of the terminals of structure Z. Which terminal (X or Y) releases neurotransmitter W? (1 mark)
- ii) Describe how the neurotransmitter W at location 1 can bring about the transmission of nerve impulses at structure Z. (2 marks)
- c) What is the significance of the process in (b) to the transmission of nerve impulses? (1 mark)

8. 2022Q4c

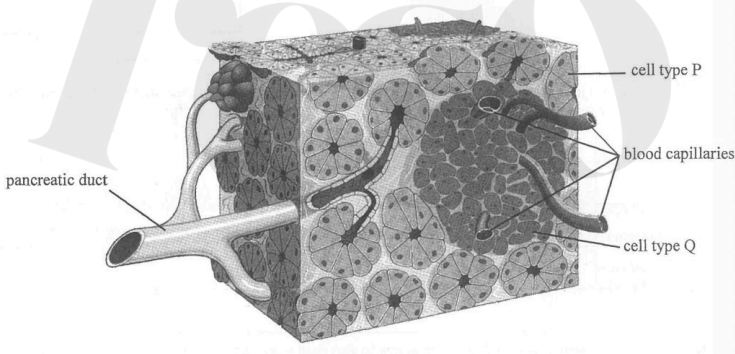
The photomicrograph below shows the karyotype of a patient who is suffering from a certain brain disease:



- c) The cerebellum is one of the regions affected by this disease. In relation to the function of the cerebellum, suggest *one* difficulty that would be experienced by this patient. (1 mark)

9. 2022Q6a

The schematic diagram below illustrates the distribution of different cell types in the human pancreas:



- a) Which type of cells, P or Q, secretes hormones? Support your answer with *one* observable feature illustrated in the diagram. (3 marks)

10. 2024Q1

Fill in the table below to compare the characteristics of nervous and hormonal controls. (3 marks)

		Nervous control	Hormonal control
(a)	Signalling molecule		
(b)	Transmission pathway		
(c)	Comparison of the time taken to induce responses		

Ans			
2012Q4: B (63%)	2012Q5: D (61%)	2012Q6: A (79%)	2015Q22: A (86%)
2015Q23: C (84%)	2017Q29: C (53%)	2017Q30: D (87%)	2018Q20: D (50%)
2018Q21: C (58%)	2018Q22: A (76%)	2019Q18: A (61%)	2019Q21: D (54%)
2020Q10: B (86%)	2023Q20: A (42%)	2024Q23: A (15%)	2024Q24: D (46%)
2024Q25: A (41%)	2024Q26: C (46%)		

2013Q1		
5	B	(1)
	C	(1)
	A	(1)
		3 marks

2014Q10b		
10	(b)	
	•	impulses received from the visual cortex will be interpreted at the association area (1)
	•	where relevant information about the image will be retrieved from the previously stored information (1) (2)

2015Q5		
5	(a)	
	•	motor neurone (1)
	•	it is connected to an effector (1), i.e. muscle fibre in this case (2)
	(b)	
	•	S provides energy (1)
	•	for the synthesis / secretion / resynthesis of the neurotransmitter / chemical messenger stored in T (1) (2)
	(c)	
	•	arrival of nerves impulses at the motor nerve ending triggers the release of neurotransmitters / chemical messengers into the neuromuscular junction (1)
	•	these neurotransmitters / chemical messengers diffuse across the synapse (1)
	•	bind to the receptor sites on the membrane of the muscle fibre (1) to trigger muscle contraction (3)
		7 marks

2016Q1		
1		
	Condition	Part of the brain
	Difficulty in breathing	medulla / brain stem (1)
	Difficulty in balance during movement / walking (1)	cerebellum
	Difficulty in speech and vision	cerebrum / cerebral cortex (1)
		(3)
		3 marks

2019Q4		
4	(a)	
	•	Q and S (1)
	(b)	
	Any <i>two</i> of the following:	
	•	insufficient number of vesicles at the synaptic knob / insufficient number of mitochondria to provide energy for vesicles to synthesize neurotransmitters (1)
	•	insufficient numbers of neurotransmitters are released at the neuromuscular junction to stimulate R (1) (2)
	•	neurotransmitters degraded too rapidly (1)
	•	insufficient number of receptors to bind with neurotransmitters (1)
		3 marks

2020Q3aii, c		
3	(a) (ii)	
	•	motor neuron(e) OR motor (1)
	(c)	
	•	it is an involuntary action / a reflex / stereotyped response / inborn action / does not involve cerebrum for making decision (1) (any <i>one</i>) (1)

2021Q2

- 2 (a) • synapse (1) (1)
- (b) (i) • terminal X (1) (1)
- (ii) • neurotransmitter W at location 1 diffuses to the membrane of terminal Y (1)
• neurotransmitter W stimulates the membrane at terminal Y / binds to the receptor on the membrane of terminal Y to initiate a nerve impulse (1) (2)
- (c) • this ensures that the neurotransmission takes place in one direction only (1) (1)
-
- 5 marks

2022Q4c

- 4 (c) Concept for mark award (1)
- | |
|--|
| • a description of an activity which is related to balance or coordination of movement (1) |
|--|
- e.g.
- activities related to balance : the patient may trip easily while walking / may experience difficulty in walking / cannot maintain body balance (1)
- OR
- activities related to coordination of movement: the patient may not be able to walk along a straight line / perform fine movement / perform movement smoothly (1)
-
- 6 marks

2022Q6a

- 6 (a) • cell type Q (1) (1)
- | | |
|---|-----|
| Concept for mark award | (2) |
| • observable features related to cell type Q (1) | |
| • deduction from the feature in relation to hormone secretion (1) | |
- e.g.
- there are several blood capillaries in close proximity to cell type Q (1)
 - where hormones will be released into the blood (1)
- OR
- there is no duct connected to cell type Q (1)
 - therefore, the hormonal secretion should be released into the blood (1)

2024Q1

	Concept for mark award	Example	Marks
1(a)	correct signalling molecules	nervous control: neurotransmitter hormonal control: hormone	(1)
1(b)	correct transmission pathways	nervous control: neurones / nerves / nerve fibres hormonal control: blood / blood vessel	(1)
1(c)	correct comparison	nervous control is faster than hormonal control	(1)

3 marks